HR ANALYTICS

Workforce Composition, Performance & Attrition

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SQL • Excel • Power Query • DAX • Power BI

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BUSINESS PROBLEMS

Key Workforce Challenges Identified

PILLAR 1 — Workforce Visibility

- Limited real-time visibility into workforce size and composition.
- Weak insight into hiring vs attrition movement.

PILLAR 2 — Attrition & Retention Risks

- Attrition drivers unclear.
- High-risk roles not identified.
- No early warning indicators for high-risk employees.

PILLAR 3 — Diversity, Performance & Compensation

- Diversity ratios unclear.
- Performance–attrition link not well understood.
- Compensation gaps and promotion bottlenecks.

PILLAR 4 — HR Strategy & Policy Effectiveness

- HR initiatives not measured effectively.
- Limited analytics for strategic workforce planning.

PROJECT OBJECTIVES

Strategic goals of the HR Analytics System

- Build an end-to-end HR analytics system
- Create multi-page dashboards for major HR functions
- ⚠ Identify high-risk employees and their drivers
- Enable data-driven HR intervention planning
- Reduce attrition and improve workforce stability

DATA SOURCES & SQL PIPELINE

Employee Master CSV (Raw Source Data)

Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	Gender	HourlyRate	JobInvolvement	JobLevel	JobRole	JobSatisfaction
41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	2	Female	94	3	2	Sales Executive	4
49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	3	Male	61	2	2	Research Scientist	2
37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	4	Male	92	2	1	Laboratory Technician	3
33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	4	Female	56	3	1	Research Scientist	3
27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	1	Male	40	3	1	Laboratory Technician	2
32	No	Travel_Frequently	1005	Research & Development	2	2	Life Sciences	1	8	4	Male	79	3	1	Laboratory Technician	4
59	No	Travel_Rarely	1324	Research & Development	3	3	Medical	1	10	3	Female	81	4	1	Laboratory Technician	1
30	No	Travel_Rarely	1358	Research & Development	24	1	Life Sciences	1	11	4	Male	67	3	1	Laboratory Technician	3
38	No	Travel_Frequently	216	Research & Development	23	3	Life Sciences	1	12	4	Male	44	2	3	Manufacturing Director	3
36	No	Travel_Rarely	1299	Research & Development	27	3	Medical	1	13	3	Male	94	3	2	Healthcare Representative	3
35	No	Travel_Rarely	809	Research & Development	16	3	Medical	1	14	1	Male	84	4	1	Laboratory Technician	2
29	No	Travel_Rarely	153	Research & Development	15	2	Life Sciences	1	15	4	Female	49	2	2	Laboratory Technician	3
31	No	Travel_Rarely	670	Research & Development	26	1	Life Sciences	1	16	1	Male	31	3	1	Research Scientist	3
34	No	Travel_Rarely	1346	Research & Development	19	2	Medical	1	18	2	Male	93	3	1	Laboratory Technician	4
28	Yes	Travel_Rarely	103	Research & Development	24	3	Life Sciences	1	19	3	Male	50	2	1	Laboratory Technician	3
29	No	Travel_Rarely	1389	Research & Development	21	4	Life Sciences	1	20	2	Female	51	4	3	Manufacturing Director	1
32	No	Travel_Rarely	334	Research & Development	5	2	Life Sciences	1	21	1	Male	80	4	1	Research Scientist	2
22	No	Non-Travel	1123	Research & Development	16	2	Medical	1	22	4	Male	96	4	1	Laboratory Technician	4
53	No	Travel_Rarely	1219	Sales	2	4	Life Sciences	1	23	1	Female	78	2	4	Manager	4

MySQL Table Preview After Import

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	Gender	HourlyRate	JobInvolvement	J
41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	2	Female	94	3	2
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37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	4	Male	92	2	1
33	No	Travel_Freque Travel_Frequently		esearch & Development	3	4	Life Sciences	1	5	4	Female	56	3	1
27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	1	Male	40	3	1
32	No	Travel_Frequently	1005	Research & Development	2	2	Life Sciences	1	8	4	Male	79	3	1
59	No	Travel_Rarely	1324	Research & Development	3	3	Medical	1	10	3	Female	81	4	1

SQL Join: Combining Employee + Attrition

```sql
select *
from employee_attrition as ea
join employee_data as ed
on ea.EmployeeNumber = ed.EmployeeNumber;
```

SQL Pipeline Summary

- Raw CSV loaded into MySQL staging
- Data cleaned and standardized
- NULLs and inconsistent values removed
- Data types converted (int, text, dates)
- Tables joined to create an analysis-ready HR dataset

DATASET OVERVIEW

1.Dataset Size & Structure
- **1470 employee records**
- **35 HR attributes** per employee
- **1 row = 1 employee profile**
- **0 duplicate employee IDs**
- **100% structured tabular format**

2. Data Coverage Breakdown
- **Demographic fields** – Age, Gender, Marital Status, Education, etc.
- **Job-related fields** – Department, Job Role, Job Level, Salary, Travel, etc.
- **Tenure-related fields** – Years at Company, Years in Role, Promotion Gap, etc.
- **Satisfaction metrics** – Job, Environment, Relationship, etc.
- **Performance-related fields** – Rating, Overtime, Training, etc.
- **Attrition risk indicators** – Attrition flag, Overtime, Distance From Home, etc.

3. Employee Characteristics & Workforce Patterns
- **Average age: ~37 years**
- **Tenure range: 0–40 years**
- **Majority tenure group: 2–4 years**
- **Highest headcount roles:** Sales Executive, Research Scientist
- **Largest departments:** R&D, Sales

4. Key HR Topics Covered in Dataset
- **Demographics:** Age, Gender, Marital Status, Education
- **Workforce structure:** Department, Role, Job Level
- **Experience:** Total tenure, years in role, years since last promotion
- **Performance:** Rating, training, workload indicators
- **Engagement:** Satisfaction metrics across multiple categories
- **Risk Factors:** Attrition, overtime, distance, job satisfaction

5. Data Quality Highlights
- **Consistent formatting** across numeric & categorical fields
- **No missing employee IDs**
- **No duplicated records** found
- **Columns with categorical data fully standardized**
- **Clean numerical ranges** for age, tenure, rating, satisfaction

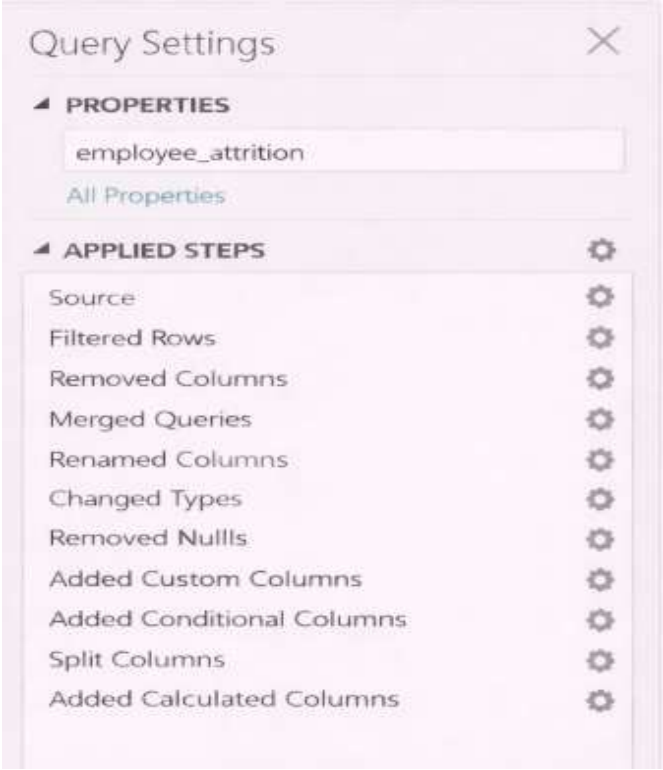
6. Why This Dataset Is Strong for HR Analytics
- **Complete employee life-cycle snapshot** in a single dataset
- **Balanced mix of qualitative & quantitative HR features**
- **Allows deep segmentation** (role, department, tenure, rating, satisfaction)
- **Ideal for attrition prediction, workforce planning & risk analysis**
- **Includes all 4 pillars:** workforce composition, performance, satisfaction, risk

7. Final Dataset Summary Box (For Your Slide Footer)
- **1470 employees**
- **35 HR attributes**
- **Covers demographics → job → tenure → performance → satisfaction → risk**
- **Clean, complete, analysis-ready dataset**

POWER QUERY & DAX LAYER

Power Query Transformations

- ✓ Source (SQL-Joined Table)
- ✓ Filtered Rows
- ✓ Removed Columns
- ✓ Merged Queries
- ✓ Renamed Columns
- ✓ Changed Types
- ✓ Removed Nulls
- ✓ Added Custom Columns
- ✓ Added Conditional Columns
- ✓ Split Columns
- ✓ Added Calculated Columns



Key DAX Measures

```
1.Total Emp = COUNTROWS('employee_attrition')

2.Avg Tenure = ROUND(AVERAGE(employee_attrition[YearsAtCompany]), 1)

3.% Due for Promotion = DIVIDE([Due for Promotion], [Total Emp], 0)

4.% Employees on Overtime =
  VAR _pct = DIVIDE([Employees on Overtime], [Total Emp], 0)
  RETURN IF(ISBLANK(_pct), 0, _pct)

5.% High Rated = DIVIDE([High Rated], [Total Emp], 0)

6.High Risk % =
  DIVIDE(
    CALCULATE(
        COUNTROWS(employee_attrition),
        employee_attrition[RiskScore] >= 7
    ),
    [Total Emp],
    0
)
```

EXECUTIVE OVERVIEW — PURPOSE & INSIGHTS

High-level snapshot of the workforce to support strategic HR decisions

WHY THIS DASHBOARD EXISTS

Provides HR leadership with a high-level snapshot of overall workforce health and key organizational indicators on a single page.

WHAT PROBLEMS IT SOLVES

- No centralized view of major workforce KPIs
- Hard to assess gender balance, tenure mix, and movement indicators
- Leadership lacked a quick summary before deeper analysis

KEY QUESTIONS ANSWERED

- How many employees do we have?
- What is the gender split?
- Who is due for promotion or retrenchment?
- How is experience and job level distributed?

KEY INSIGHTS & VALUE

- Instant visibility into workforce size and structure
- Clear view of promotion readiness and service status
- Quick understanding of tenure and job level mix

BUSINESS VALUE & IMPACT

- Enables fast, data-driven HR decisions
- Supports budgeting, manpower planning, and workforce forecasting
- Serves as the entry point for all deeper HR dashboards

WORKFORCE ANALYSIS — PURPOSE & INSIGHTS

Detailed breakdown of workforce composition, roles, levels, and experience.

WHY THIS DASHBOARD EXISTS

- Provides a detailed breakdown of workforce composition across
- departments, job roles, experience categories, job levels, and satisfaction.

WHAT PROBLEMS IT SOLVES

- No single view showing how the workforce is distributed
- Difficult to identify hiring pressure across departments
- No visibility into job role and job level imbalance

KEY QUESTIONS ANSWERED

- Which departments are the largest?
- Which roles dominate the workforce?
- How is tenure distributed across employees?
- What is the mix of job levels in the organization?
- What is the satisfaction distribution across staff?

KEY INSIGHTS & VALUE

- R&D and Sales dominate workforce headcount
- Majority of employees fall in 2–4 year experience range
- Job roles like Sales Executive and Research Scientist have high volumes
- High satisfaction levels seen in a significant portion of the workforce

BUSINESS VALUE & IMPACT

- Helps HR optimize hiring plans
- Highlights areas needing better workload allocation
- Supports leadership in understanding skill and experience mix
- Enables HR to structure workforce, identify pressure points, and plan hiring, internal movements

ATTRITION ANALYSIS — PURPOSE & INSIGHTS

Understanding retrenchment, promotion delays, satisfaction levels, and root causes

WHY THIS DASHBOARD EXISTS

Provides HR leaders with a clear view of attrition patterns, retrenchment levels, promotion delays, and satisfaction-driven risk indicators across departments and roles.

WHAT PROBLEMS IT SOLVES

- No centralized view of attrition reasons and department-wise risk
- Hard to identify teams with high retrenchment or promotion backlog
- No quick way to analyze satisfaction levels linked to attrition
- Leadership lacked early signals of workforce instability

KEY QUESTIONS ANSWERED

- Which departments have the highest retrenchment counts?
- How many employees are overdue for promotion?
- What are the primary reasons for attrition?
- Are employees satisfied or at risk?
- Which roles or departments require immediate attention?

KEY INSIGHTS & VALUE

- Sales and R&D show the highest retrenchment volumes
- Promotion delays visible in key roles (Manager, Technician, Scientist)
- “Underperforming” and “Reorg” are the top attrition reasons
- High-rated employees dominate, but low-rated groups reveal hidden risks

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BUSINESS VALUE & IMPACT

- Enables HR to target problem areas with data-backed intervention
- Helps plan promotion cycles and reduce retrenchment risk
- Supports organizational stability through early identification of at-risk groups
- Drives better retention strategy by linking satisfaction to attrition outcomes

DRIVERS & RISK ANALYSIS — PURPOSE & INSIGHTS

Identify key factors driving attrition and highlight high-risk employee segments.

WHY THIS DASHBOARD EXISTS

Provides HR with insights into the root causes of attrition and identifies employees at highest risk due to dissatisfaction, overtime, job stagnation, or short tenure.

WHAT PROBLEMS IT SOLVES

- No clear visibility into WHY employees are at risk
- Hard to detect behavioral & performance drivers behind attrition
- Difficult to compare risk segments across roles, levels, and departments
- No early warning indicators for high-risk employees

KEY QUESTIONS ANSWERED

- What factors contribute most to employee dissatisfaction?
- Which segments show the highest attrition risk?
- Which roles/departments have the highest-risk employees?
- How does job satisfaction, overtime, or tenure influence attrition?
- Who are the employees most likely to leave?

KEY INSIGHTS & VALUE

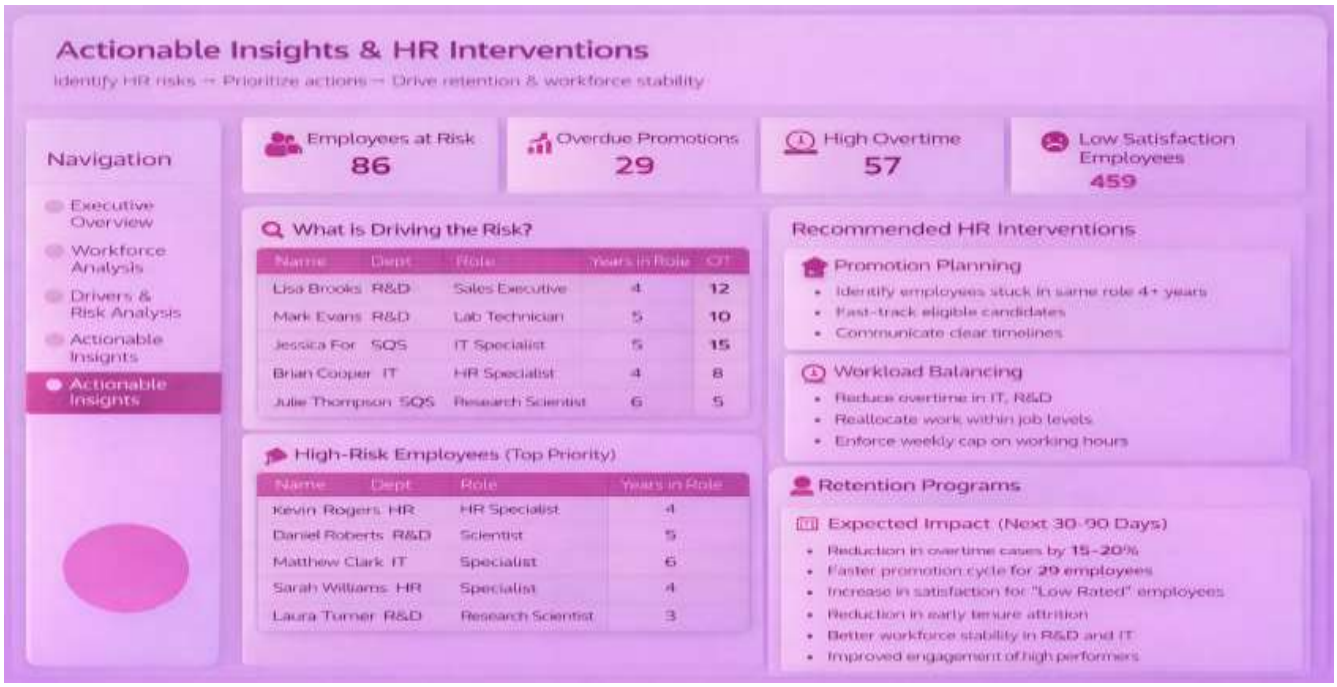
- Overtime workers and dissatisfied employees are top risk segments
- Short tenure (<2 years) significantly increases attrition likelihood
- Specific roles (Sales Exec, Lab Technician, Research Scientist) show elevated risk
- R&D and Sales departments contain the highest-risk groups
- High-risk employees are highly concentrated in low-satisfaction & high-overtime clusters

BUSINESS VALUE & IMPACT

- Enables accurate targeting of high-risk employee groups
- Supports proactive decision-making with early attrition warning signals
- Reduces turnover cost by addressing root causes rather than symptoms
- Strengthens workforce stability and reduces future attrition rates

ACTIONABLE INSIGHTS & HR INTERVENTIONS

Identify HR risks → Prioritize actions → Drive retention & workforce stability



WHY THIS DASHBOARD EXISTS

Provides a prioritized view of employees needing urgent HR action, enabling leadership to address overtime, promotion delays, dissatisfaction, and role stagnation quickly and effectively.

WHAT PROBLEMS IT SOLVES

- No centralized list of high-risk employees
- Hard to identify root causes behind employee risk
- No structured way to prioritize HR actions
- Limited visibility into promotion delays & workload issues
- Lack of early warning for retention risks

KEY QUESTIONS ANSWERED

- Who are the highest-risk employees?
- What factors are driving their risk?
- Which HR actions (promotion, workload balance, retention) are needed?
- Who should be prioritized first?
- What impact will interventions create?

KEY INSIGHTS & VALUE

- 86 employees identified as high-risk
- 29 employees overdue for promotion
- Overtime (57) & low satisfaction (459) are major drivers
- Core risk factors: dissatisfaction, overtime, stagnation, role mismatch
- High-risk employees clearly mapped by name, role, tenure & department

BUSINESS VALUE & IMPACT

- Enables timely, targeted HR interventions
- Improves retention by addressing root causes
- Reduces overtime & workload imbalance
- Accelerates promotion cycles
- Creates measurable impact within 30–90 days
- Strengthens overall workforce stability

FINAL KEY INSIGHTS & RECOMMENDATIONS

Summary of Workforce Health, Risks & Strategic HR Actions

STRATEGIC WORKFORCE INSIGHTS

KEY WORKFORCE FINDINGS

- Total workforce: 1,470 employees
- R&D and Sales form the largest employee groups
- Majority tenure between 2–4 years
- Healthy job level distribution
- High proportion of "High Rated" performers

KEY RISKS IDENTIFIED

KEY RISK OBSERVATIONS

- 72 employees overdue for promotion
- 117 employees flagged for retrenchment
- Overtime risk: 57 employees
- Low satisfaction: 459 employees
- Short tenure + stagnation driving majority risks

RECOMMENDED HR ACTIONS (Promotion • Workload • Retention)

RECOMMENDED HR ACTIONS

PROMOTION PLANNING

- Fast-track employees with 4+ years in same role
- Clear criteria & timelines for 29 overdue promotions
- Improve career growth visibility

WORKLOAD BALANCING

- Reduce overtime in R&D, IT and Sales
- Redistribute workload within job levels
- Implement clear limits on weekly working hours

RETENTION PROGRAMS

- Target dissatisfaction with focused interventions
- Enhance recognition for high performers
- Provide career development for mid-tenure employees

EXPECTED BUSINESS IMPACT (Next 30–90 Days Projection)

BUSINESS IMPACT (Next 30–90 Days)

- Overtime reduced by 15–20%
- Promotion cycle accelerated for 29 employees
- Drop in early-tenure attrition
- Improved satisfaction among low-rated employees
- Stable workforce levels in R&D & Sales
- Stronger engagement from high performers